Madhav Venkatachalam

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Education

Cornell University Ithaca, NY

B.S. in Applied and Engineering Physics, Minor in Mathematics *May 2029*

- **GPA:** 4.03/4.00
- **Relevant Coursework:** Calculus 3, Differential Equations, Linear Algebra, Real Analysis, Number Theory, Honors Mechanics and Special Relativity, Honors Electricity and Magnetism, Honors Waves and Oscillations, Experimental Physics 1&2, Lasers and Photonics, Quantum Information and Hardware
- **TA for AEP 1100: Lasers and Photonics**

Experience

Fuchs Group (Cornell University) September 2025 – Present

Undergraduate Researcher *Ithaca, NY*

- Simulated high-throughput microwave packages for precise superconducting device measurement using COMSOL and Ansys; Designed a modified package using Autodesk Fusion to minimize parasitic modes
- Tested optical modes in SiC waveguides and ring resonators using a tunable cavity coupling to spin defects
- Designed and fabricated a novel PCB and backing plate to better house multiple 10mm superconducting chips

Massachusetts Institute of Technology July 2026 – August 2026

Teaching Assistant *Remote*

- Taught MIT Beaver Works Summer Institute's Quantum Software course alongside other teaching staff
- Covered quantum algorithms such as Deutsch-Jozsa, Bernstein-Vazirani, Simon's, Grover's, and Shor's
- Guided final projects (e.g. Variational Quantum Eigensolver and QAOA for Vehicle Routing) in Qiskit and Q#

Idlexperts/Idler (YCS25) March 2026 – Present

Problem Design Engineer *Remote*

- Evaluated agent-generated failure analyses for Idler's RL training pipeline, and built environments for testing
- Reviewed production-level code changes across repos to evaluate problem clarity and fairness for training datasets

Katz Lab (Cornell University) December 2025 – May 2026

Undergraduate Researcher *Ithaca, NY*

- Designed low-pass RLC filters for coherent quantum entanglement of a Barium trapped ion quantum computer
- Built a temperature monitoring system for a laser-cooled optical ion trap. Engineered optical housing for the trap

Extracurriculars & Projects

Quantum Counterfeit Coin Solver | *Python, Qiskit 2.6, Qiskit Aer* October 2026

- Built working circuit for Iwama et al.'s $O(k^{1/4})$ query quantum algorithm for finding all k counterfeit coins, including a reversible balance oracle and a coherent, superposition-controlled state-preparation circuit
- Wrote an exact sparse quantum simulator (matches Qiskit Aer to 10^{-15}) and a 208-test suite, and designed a schedule optimization that cut oracle queries from 9 to 5 and gate count by ∼40% at a fixed 99% success target

Cornell Hyperloop | *Magentics Subteam* September 2026 – Present

- Utilized Ansys Maxwell to design a linear induction motor's cooling system based on simulated heat generation
- Simulated magnetic pod response and drive on MATLAB to adjust sensors and controls of the levitation system
- Designed and built an electromagnet characterization rig to quantify existing electromagnetic drives

Quantum Computing Association | *Theory and Decoding Subteam* September 2025 – Present

- Theory Subteam: Worked on developing quantum low density parity check codes through Z-X calculus
- Decoding Subteam: Worked on optimizing Union Find algorithm and Blossom algorithm for MWPM

Cornell Venture Capital | *Analyst* February 2026 – Present

- Worked with Insight Partners and Foundation Capital to deliver analysis and early-stage investment opportunities
- Focused on market analysis of the Hardware layer of the physical intelligence and robotics industries

Skills

Simulation and Engineering/Physics: COMSOL, Ansys, LT Spice, Fusion, AutoCAD, KLayout, KiCad
Languages: Java, JavaScript C\C++, Q#, Python, MATLAB, HTML\CSS, Rust, SQL, Verilog
Libraries and Tools: Pandas, NumPy, Matplotlib, Node.js, Qiskit, Qiskit Aer, Stim, Git, VS Code, Jupyter, Cursor